

Adversarial attacks and defenses in EEG based Brain Computer Interfaces: A comprehensive survey and future directions

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ABSTRACT

Brain–Computer Interfaces (BCIs), particularly those based on electroencephalography (EEG), have demonstrated significant potential in assistive technologies, neurorehabilitation, and human–machine interaction. However, their reliance on machine learning models makes them vulnerable to adversarial exploitation. This paper aims to systematically investigate adversarial attacks targeting EEG-based BCI systems and to analyze their implications for reliability and security. Specifically, we examine representative attack scenarios, including adversarial example generation through gradient-based perturbations (e.g., FGSM and PGD), data poisoning during model training, and physical signal perturbations affecting EEG acquisition. These attacks can induce misclassification, degrade system performance, or manipulate user intent decoding without noticeable changes in the signal. By characterizing the objectives, threat models, and impact of such attacks across different stages of the BCI pipeline, this study identifies critical vulnerabilities inherent in EEG-based systems. Furthermore, we review existing defense strategies and highlight current limitations, thereby outlining research directions toward developing robust and secure EEG-based BCI frameworks resilient to adversarial exploitation.

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1. Introduction

Brain–Computer Interfaces (BCIs) represent a transformative technology that enables direct communication between neural activity and external devices [1]. By translating brain signals into executable commands, BCIs establish alternative communication pathways that bypass conventional neuromuscular channels. This capability is particularly impactful in healthcare, where BCIs support motor restoration, neurorehabilitation, and assistive technologies for individuals with neurological impairments. Representative applications include wheelchair navigation [2], robotic arm control [3], speller systems [4], and neurofeedback training [5]. Beyond clinical contexts, BCIs are increasingly explored in entertainment [6], gaming [7], military systems [8], and advanced human–computer interaction paradigms [9], reflecting their expanding societal relevance [10].

Among available neural acquisition modalities, electroencephalography (EEG) remains the most widely adopted in practical BCI systems due to its non-invasive nature, portability, relatively low cost, and

high temporal resolution [11,12]. Nevertheless, EEG signals are characterized by a low signal-to-noise ratio (SNR), non-stationarity, inter-subject variability, and susceptibility to physiological and environmental artifacts [13]. These intrinsic challenges necessitate advanced computational methods capable of extracting robust and discriminative representations from complex neural data. Recent advances in deep learning have significantly enhanced EEG-based BCI research by enabling end-to-end representation learning [14]. Unlike traditional pipelines that rely on handcrafted features (e.g., common spatial patterns) combined with classical classifiers such as linear discriminant analysis (LDA) or support vector machines (SVM), deep neural networks automatically learn hierarchical spatial–temporal features directly from raw or minimally processed EEG signals. Convolutional neural networks (CNNs), in particular, effectively model temporal dynamics, spatial channel correlations, and spectral characteristics. Architectures such as EEGNet [15] and Deep/Shallow ConvNets [16] have become strong baselines across motor imagery, event-related potential, and P300 paradigms. These developments have improved accuracy, adaptability, and real-time deployment capabilities of modern BCIs.

Despite these advances, the integration of deep learning has introduced new security vulnerabilities. Deep neural networks are known

to be susceptible to adversarial attacks, in which carefully crafted and often imperceptible perturbations are added to input data to induce misclassification [17]. Initially studied in computer vision, adversarial machine learning has since expanded to speech processing [18], natural language processing [19], and increasingly to neural signal decoding. In EEG-based BCIs, adversarial perturbations may lead to arbitrary or targeted misclassification, intent manipulation, or unsafe device behavior. For example, an attacker could cause a speller system to output incorrect characters, alter neurofeedback signals during rehabilitation, or manipulate control commands in brain-driven wheelchairs or prosthetic devices.

Empirical studies have demonstrated the feasibility of multiple attack vectors in EEG-based BCIs. Gradient-based evasion methods such as the Fast Gradient Sign Method (FGSM) [20] and Projected Gradient Descent (PGD) have been shown to significantly degrade classifier performance [21,22]. Universal adversarial perturbations reveal that a single crafted distortion can generalize across trials or subjects. Sparse perturbation strategies demonstrate that modifying only a small subset of EEG samples can remain effective. Moreover, black-box attacks – where the adversary has no access to model parameters – have proven viable through surrogate modeling and query-based strategies. Frequency-domain attacks further exploit spectral properties of EEG signals, while poisoning attacks target the training phase by injecting malicious samples into calibration datasets. Collectively, these findings indicate that adversarial threats in BCIs are both technically feasible and can potentially have a high impact.

Although adversarial machine learning is well established in computer vision [23,24] and natural language processing [19], EEG-based BCIs present distinct challenges. First, EEG signals exhibit low SNR, high variability, and session-dependent distribution shifts. Second, perturbations may exploit neurophysiological characteristics such as band-specific power variations or spatial channel dependencies. Third, many BCIs operate in real-time closed-loop environments, increasing the risk of immediate safety-critical consequences. Finally, ethical and regulatory implications are amplified because BCIs directly interface with human cognitive processes. These characteristics make EEG-based BCIs a uniquely vulnerable yet comparatively underexplored domain for adversarial robustness research.

1.1. Research gap and survey contributions

While prior surveys have addressed general security and privacy issues in BCIs [25–27], none provides a dedicated and systematic synthesis of adversarial machine learning threats and defenses specific to EEG-based BCI systems. Existing reviews often treat adversarial risks as a peripheral concern, lack unified threat modeling, and do not critically evaluate robustness assessment methods or defense effectiveness. Furthermore, standardized evaluation benchmarks and real-world attack analyses remain limited.

To address these gaps, this survey provides a comprehensive and structured analysis of adversarial attacks and defense mechanisms in EEG-based BCIs. Our primary objective is to systematically examine how adversarial exploitation can compromise neural decoding pipelines and to identify principled strategies for enhancing robustness, reliability, and safety.

The main contributions of this survey are fourfold:

- (1) **Unified Taxonomy:** We introduce a structured taxonomy for adversarial attacks in the BCI context, categorizing them along multiple dimensions including attack stage (evasion vs. poisoning), adversary knowledge (white-, gray-, black-box), specificity (targeted vs. non-targeted), and perturbation frequency.
- (2) **Systematic Analysis of Adversarial Defenses:** This work offers the first systematic classification and critical evaluation of adversarial defense mechanisms tailored for EEG based BCIs. We organize defenses into three primary categories—detection-based, input-improvement, and model-improvement strategies—analyzing their principles, efficacy, and limitations.

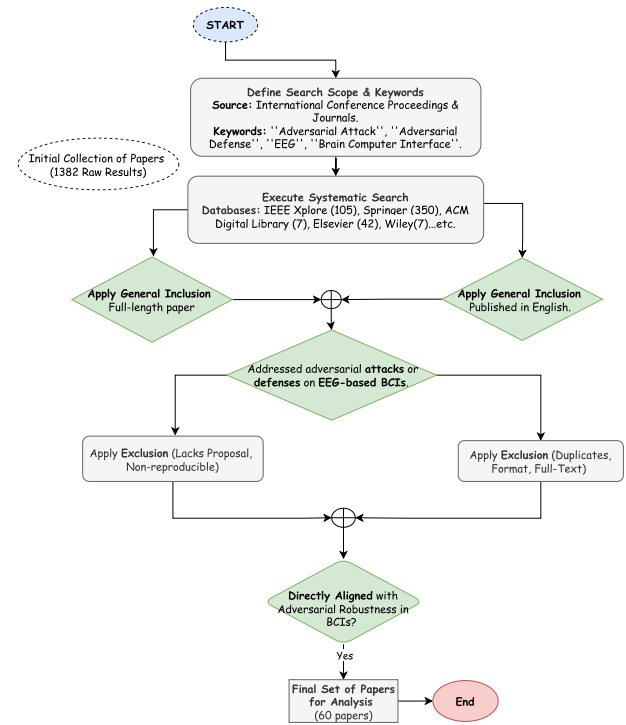


Fig. 1. Selection process for survey papers (PRISMA-based flow diagram).

- (3) **State-of-the-Art Consolidation:** This survey consolidates research published up to 2025, capturing the most recent advancements, including universal adversarial perturbations, frequency-domain attacks, generative adversarial network (GAN)-based threats, and emerging robust training paradigms that have not been comprehensively covered in prior reviews.
- (4) **Identification of Critical Research Gaps:** We highlight key unresolved issues, including the absence of standardized robustness benchmarks, limited real time and human in the loop evaluations, robustness–accuracy–efficiency trade-offs, certified robustness for neural signals, and regulatory considerations in safety-critical deployments.

1.2. Selection process for survey papers

To ensure methodological transparency and reproducibility, this review follows a structured screening protocol inspired by the PRISMA-ScR framework. Relevant studies were identified through systematic searches using the keywords “adversarial attack”, “adversarial defense”, “EEG”, and “brain–computer interface” across major scientific databases, including Scopus, Web of Science, IEEE Xplore, Springer, ACM Digital Library, Elsevier, and Google Scholar.

Inclusion criteria comprised full-length peer-reviewed journal and conference papers published in English that explicitly propose or evaluate adversarial attacks or defenses in EEG-based BCI systems. Exclusion criteria eliminated duplicate entries, short papers (fewer than two pages), white papers, studies lacking reproducible experimental validation, and works not evaluated on EEG or BCI datasets. Titles, abstracts, introductions, and conclusions were screened to ensure direct relevance to adversarial robustness in neural decoding. The overall selection workflow is illustrated in Fig. 1.

1.3. Organization of the paper

The general organization of this paper is as follows: Section 2 presents an overview of Brain–Computer Interface systems, detailing

their system architecture and core components. Section 3 reviews machine learning and deep learning classifiers for EEG decoding, ranging from traditional feature-based methods to advanced neural network architectures. Section 4 introduces the adversarial attack and defense framework, covering formal formulations, taxonomies, and methodological foundations. Section 5 explains how BCI systems can be compromised by adversarial manipulation within the signal-processing and classification pipeline. Section 6 provides a comprehensive analysis of adversarial attack strategies in BCI systems. Section 7 discusses adversarial defense mechanisms designed to enhance BCI robustness. Section 8 focuses on adversarial robustness evaluation, including benchmark datasets, experimental protocols, and performance metrics. Section 9 highlights current research challenges and open issues while outlining potential future directions. Finally, Section 10 concludes the paper by summarizing key insights and reflecting on the development of resilient and secure EEG-based BCI systems.

2. Brain-computer interface (BCI): Overview and system architecture

This section discusses the background knowledge and working algorithm involved in developing BCIs. It provides the foundation and basic knowledge necessary for understanding the complexities of BCI systems. Additionally, it reviews previous research efforts that have laid the groundwork for current advancements in this field. By presenting this information, this section aims to offer readers a comprehensive overview and better understanding of the context in which BCIs are being developed and improved.

2.1. Historical overview

The evolution of BCI spans over a century, progressing from foundational neurophysiological discoveries to the development of advanced neuroprosthetics.

2.1.1. Foundational developments

The field was significantly advanced by Hans Berger, who pioneered EEG in the 1920s. His work led to the first recording of human brainwaves in 1924, which notably included the identification of alpha rhythms. This non-invasive technique laid the essential groundwork for subsequent neural signal detection. By the 1960s, practical applications began to emerge, such as Alvin Lucier's use of EEG signals to generate musical compositions.

2.1.2. Emergence of BCI research

The formal concept of a "BCI" was articulated by Jacques Vidal in 1973 at the University of California, Los Angeles (UCLA), with support from the National Science Foundation (NSF) and the Defense Advanced Research Projects Agency (DARPA) [28]. Prior to this, foundational work by Eberhard Fetz in 1969 demonstrated that monkeys could learn to control a meter through operant conditioning of individual neuronal firing rates. Throughout the 1970s, UCLA conducted pioneering trials involving human subjects navigating virtual mazes using EEG-based cursor control.

2.1.3. Pioneering implants and trials

The 1980s marked the advent of invasive BCIs, exemplified by Phillip Kennedy's development and implantation of intracortical electrodes in monkeys in 1987. Human trials began in the mid-1990s, leveraging the principle of neuroplasticity to control neuroprosthetic devices. By 1998, neural implants enabled individuals with locked-in syndrome to perform basic device manipulation.

2.1.4. Major technological milestones

The early 2000s witnessed significant progress in decoding motor intentions from neural activity in primates during joystick tasks. This led to landmark achievements with the BrainGate initiative, which allowed quadriplegic individuals, most notably Matt Nagle, to control computer cursors and robotic limbs through implants in the motor cortex. Subsequent innovations have included the restoration of speech synthesis via BCIs in 2019 and the development of high-density electrode arrays by companies like Neuralink.

2.1.5. Contemporary advances

Ongoing refinements in signal processing, device miniaturization, and machine learning are now enabling bidirectional interfaces and increasingly sophisticated prosthetic limbs. The continued evolution of BCI technology is accompanied by important considerations regarding ethics, clinical applications, and defense-related interests.

Fig. 2 illustrates key milestones in BCI development.

2.2. BCI system taxonomy

Brain-Computer Interfaces can be categorized along several dimensions depending on how the system operates, how brain signals are recorded, and what type of neural activity is used. Fig. 3 illustrates a comprehensive taxonomy of BCI systems, showing four main classification criteria: dependability, recording method, mode of operation, and signal type. This taxonomy helps to understand the diversity of BCI paradigms, their design considerations, and their appropriate application domains.

2.2.1. According to dependability

Dependability defines whether the BCI system depends on the user's residual motor control or operates entirely through brain signals.

- *Dependent BCIs:* These systems require some muscular or peripheral activity to generate signals that can be recorded. They are typically hybrid systems combining EEG or EMG signals and are suitable for users with partial motor capabilities. For example, systems that rely on eye movement or facial muscle activity fall under this category.
- *Independent BCIs:* These systems rely solely on neural activity and are independent of any muscle movement. Independent BCIs are essential for users with severe motor disabilities, such as patients suffering from amyotrophic lateral sclerosis (ALS) or spinal cord injuries, where voluntary muscular control is lost. They are more challenging to design due to the low signal-to-noise ratio of purely neural data.

2.2.2. According to recording method

BCIs can also be distinguished by how brain activity is recorded, which directly affects the quality, safety, and applicability of the system.

- *Invasive BCIs:* These involve surgical implantation of electrodes either directly into the cortex or on its surface. Because the electrodes are in close proximity to neurons, invasive BCIs provide high spatial and temporal resolution. Examples include intracortical microelectrodes and Electrocorticography (ECoG). These systems are mainly used in medical and clinical research, such as prosthetic control and restoration of movement.
- *Non-invasive BCIs:* These record neural activity from outside the skull using sensors placed on the scalp or around the head. They are safe and more practical for everyday use, though they offer lower spatial resolution and are more prone to noise. Common non-invasive methods include Electroencephalography (EEG), Magnetoencephalography (MEG), and Functional Magnetic Resonance Imaging (fMRI). EEG based BCIs are the most popular due to their portability, cost-effectiveness, and high temporal resolution.

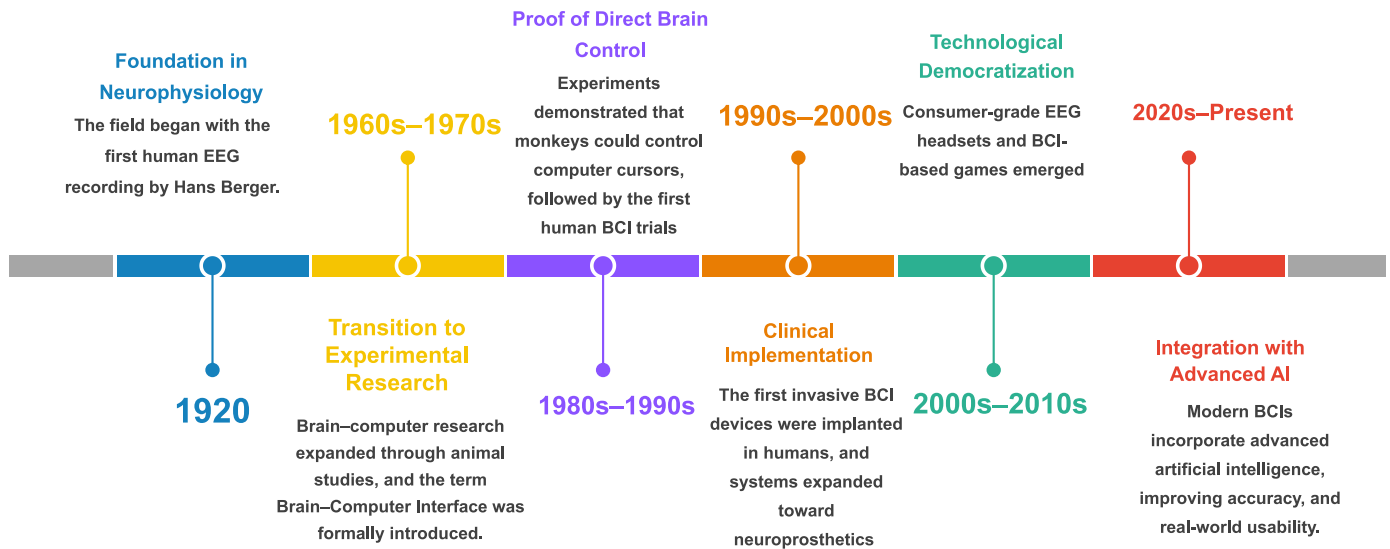


Fig. 2. Evolution of BCI technology: from early EEG discoveries to modern clinical and consumer applications.

2.2.3. According to mode of operation

The mode of operation determines how and when the user interacts with the BCI system. Two primary modes exist depending on the temporal relationship between user intent and system response.

- **Synchronous BCIs:** These systems operate within predefined time intervals, typically cued by external stimuli. The user is instructed to perform specific mental tasks during a given time window, making the system easier to synchronize and train. However, this mode restricts user autonomy and is less natural for continuous control tasks.
- **Asynchronous BCIs:** Also known as self-paced BCIs, they allow the user to issue commands at any time without external prompting. This design is more flexible and closer to natural interaction, but it is more complex to implement since the system must accurately detect voluntary commands amid spontaneous brain activity.

2.2.4. According to signal type

Brain signals used in BCIs are typically divided into two categories: evoked potentials and spontaneous activity. This classification depends on whether the brain activity is externally stimulated or self-generated.

- **Evoked Potential-Based BCIs:** These depend on neural responses elicited by external sensory stimuli such as flashes or sounds. They are highly reliable and easy to detect because the response occurs at predictable times after stimulation. Major types include:
 - **Event-Related Potentials (ERP):** Transient voltage changes in EEG associated with sensory or cognitive events. The P300 potential is a common ERP component used in communication BCIs (e.g., P300 speller).
 - **Steady-State Visually Evoked Potentials (SSVEP):** Continuous oscillatory responses induced by repetitive visual stimulation at specific frequencies. SSVEP-based BCIs are known for their high information transfer rate and minimal training requirements.
 - **P300 Responses:** Positive deflections in EEG occurring roughly 300 ms after a target stimulus, widely used in user-interface selection tasks.
- **Spontaneous BCIs:** These systems rely on brain activity that occurs naturally without external stimulation. The user voluntarily modulates their brain signals through mental tasks such as imagination or concentration. Major paradigms include:

- **Motor Imagery (MI):** Involves the imagination of body movements (e.g., hand, foot, or tongue). MI produces characteristic oscillatory patterns in the sensorimotor cortex, particularly in the mu (8–13 Hz) and beta (13–30 Hz) frequency bands.
- **Non-Motor Imagery:** Mental tasks unrelated to movement, such as arithmetic calculation or spatial rotation, can also generate distinguishable EEG patterns suitable for BCI control.

2.3. BCI systems and components

A typical BCI system consists of five main components: signal acquisition, preprocessing, feature extraction, classification (translation), and output control. In some systems, an additional feedback module is included to close the control loop (see Fig. 4) [26].

- (1) **Signal Acquisition:** Capturing brain activity through electrodes or sensors.
- (2) **Preprocessing:** Removing artifacts and enhancing signal-to-noise ratio via spectral, spatial, or artifact rejection filters.
- (3) **Feature Extraction:** Transforming raw signals into informative representations. Time-domain (e.g., amplitude, variance), frequency-domain (e.g., power spectra), and time–frequency methods (e.g., wavelets) are widely used.
- (4) **Classification:** Machine learning models map extracted features to user intentions. Traditional approaches (SVMs, LDA) are increasingly replaced by deep learning models, which can capture complex spatiotemporal dynamics.
- (5) **Feedback and Device Control:** The decoded outputs are presented to the user or used to control external devices, closing the BCI loop. Feedback may be visual, auditory, or haptic.

3. Machine learning and deep learning classifiers for EEG

Classification methods play a central role in EEG based BCI systems, as they translate extracted features into meaningful predictions of tasks.

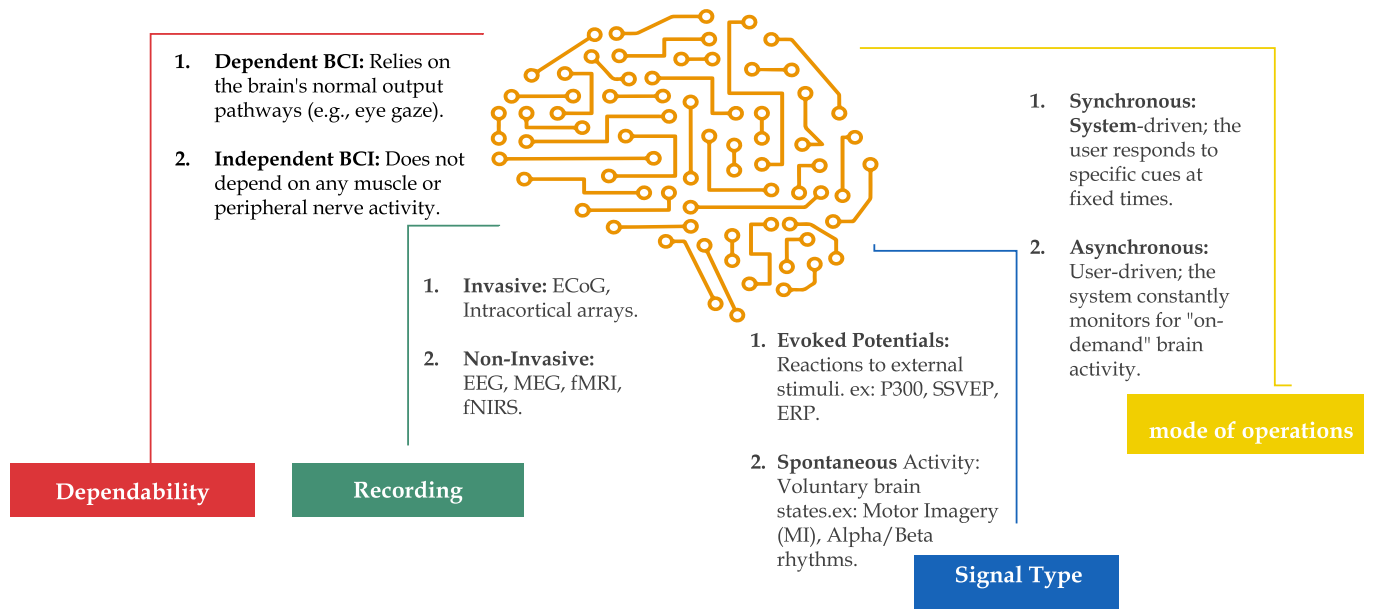


Fig. 3. Comprehensive taxonomy of BCI systems. This diagram categorizes BCIs along four primary dimensions: (1) dependability (dependent vs. independent systems), (2) recording method (invasive vs. non-invasive techniques such as EEG, MEG, ECoG, and fMRI), (3) operational mode (synchronous vs. asynchronous), and (4) signal type (evoked potentials such as P300, ERP, and SSVEP vs. spontaneous activity such as motor imagery).

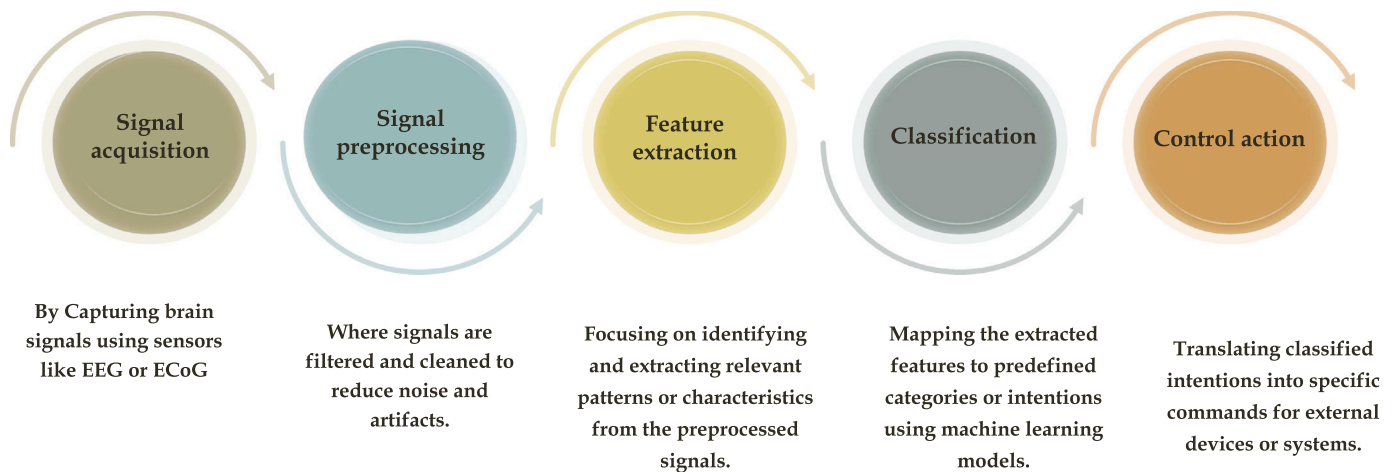


Fig. 4. BCI pipeline, illustrating major system components from neural signal acquisition to device control.

3.1. Traditional machine learning approaches

Early BCI systems relied heavily on traditional machine learning models. Linear classifiers such as Linear Discriminant Analysis (LDA) [29,30] and Logistic Regression (LR) [31,32] are widely adopted due to their computational efficiency, interpretability, and relatively strong performance on low-dimensional feature sets. However, their linear nature limits their ability to capture the complex, non-linear dynamics of EEG signals. Non-linear models, including Support Vector Machines (SVMs) [33], k-Nearest Neighbors (k-NN) [34], and Decision Trees (DTs) [35], provide better flexibility by modeling non-linear boundaries in the feature space. While effective, they typically require careful parameter tuning and are prone to overfitting when applied to high-dimensional or noisy EEG data. Probabilistic classifiers such as Naïve Bayes (NB) [36] and Quadratic Discriminant Analysis (QDA) [37] offer probabilistic decision-making frameworks, but they often rely on assumptions (e.g., feature independence) that are rarely satisfied in practice. Ensemble learning techniques, including Bagging and Adaboost [38,39], enhance predictive power by aggregating multiple

weak learners, but they come at the cost of increased complexity and computational burden. Despite their contributions, traditional ML approaches exhibit several limitations:

- (1) **Dependence on manual feature engineering:** Extracted features (e.g., CSP, PSD) require domain expertise and often fail to generalize across subjects or sessions due to EEG non-stationarity.
- (2) **Loss of temporal information:** Many classifiers operate on static feature vectors, neglecting the evolving dynamics of sensorimotor rhythms.
- (3) **Limited robustness to noise:** Physiological artifacts such as eye blinks or muscle activity can significantly degrade classification accuracy, especially in CSP-based pipelines.

3.2. Deep learning approaches

The advent of deep learning has significantly transformed EEG based BCIs by reducing reliance on handcrafted features and enabling

end-to-end learning. Convolutional Neural Networks (CNNs) such as DeepConvNet [16], ShallowConvNet [16], EEGNet [15], and EEG Inception [40] automatically capture spatial and temporal dependencies in EEG data, achieving strong performance on MI tasks. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRUs) [41,42], are designed to model temporal dynamics, addressing the sequential nature of EEG signals. Emerging paradigms such as Transformers [43,44] leverage attention mechanisms to identify and prioritize relevant features without sequential processing, offering a powerful alternative for EEG classification. Hybrid architectures (e.g., CNN-LSTM, CNN-Transformer) combine the strengths of different networks to capture both spatial and temporal dynamics more effectively [45,46]. Transfer learning (TL) [47–49] has also gained traction as a solution to data scarcity, enabling models pre-trained on large datasets to be fine-tuned for specific EEG tasks. However, these approaches often demand large volumes of high-quality labeled data and remain sensitive to inter-subject variability. Studies have shown that fixed hyperparameter choices and pre-processing strategies can significantly influence performance, limiting generalization across individuals and recording sessions.

Table 1 provides a comprehensive summary of recent deep learning approaches applied to EEG classification. The results highlight several important trends. CNN-based methods remain dominant in EEG research, achieving competitive accuracies on benchmark datasets. Notably, approaches leveraging Transfer Learning have reported higher generalization capabilities, with some studies reaching accuracies above 95% on cross-subject evaluations. Traditional DNNs, while simpler in design, tend to underperform compared to CNNs and TL-based methods, especially on more complex multi-class tasks. More recent architectures, such as LSTM autoencoders, capsule networks, and transformer-based classifiers, demonstrate promising improvements by capturing temporal dynamics, hierarchical structures, and subject-invariant representations of EEG signals.

4. Adversarial attacks and defenses: Formulation, taxonomy, and methodological overview

The following section presents a systematic investigation into the domain of adversarial attacks and defenses. It delineates a taxonomy of attack strategies, explores the operational mechanisms that facilitate their success, and evaluates contemporary defensive frameworks. A nuanced comprehension of these adversarial phenomena is critical for advancing the development of robust and secure machine learning architectures capable of maintaining functional integrity under adversarial conditions.

4.1. Theoretical formulation and motivation

Adversarial attacks represent one of the most significant vulnerabilities in modern machine learning systems. Consider a data point $x_0 \in \mathbb{R}^d$ belonging to class C_i . An adversarial attack deliberately perturbs x_0 to produce a new input x that the classifier misclassifies into another class. Goodfellow et al. (2014) [20] demonstrated that even imperceptibly small perturbations can cause highly accurate deep neural networks to make incorrect predictions, exposing the fragility of these models to subtle input manipulations. This discovery led to important questions regarding the robustness, generalization, and security of machine learning algorithms.

Although adversarial vulnerability was initially observed in deep neural networks, it is not exclusive to them. This phenomenon is an inherent property of most classifiers, stemming from the geometry of their decision boundaries in high-dimensional spaces. To better understand the underlying principles, it is essential to formally define adversarial attacks and examine their mathematical structure.

4.1.1. Formal definition and attack model

Formally, an adversarial attack seeks to transform an input $x_0 \in \mathbb{R}^d$ into a perturbed version $x \in \mathbb{R}^d$ that changes the model's predicted label. Let C_i denote the original class and C_t the target adversarial class. The goal of the attacker is to find a mapping:

$$A : \mathbb{R}^d \rightarrow \mathbb{R}^d \quad (1)$$

such that:

$$x = A(x_0) \quad (2)$$

and the classifier assigns x to C_t .

Definition 1 (Adversarial Attack). Given a data sample x_0 from class C_i and a target class C_t , an adversarial attack is a transformation $A(\cdot)$ that produces a perturbed example $x = A(x_0)$ which is misclassified as C_t . Among the various types of adversarial manipulations, the additive perturbation model is the most common due to its analytical simplicity and interpretability.

Definition 2 (Additive Adversarial Attack). An additive adversarial attack introduces a perturbation vector $r \in \mathbb{R}^d$ to the original input:

$$x = x_0 + r \quad (3)$$

where r is specifically designed to cause a misclassification into the target class C_t .

The additive model is widely used because it preserves the dimensional structure of the input (for instance, an image remains an image, or an EEG signal remains a valid signal). Additionally, it allows for direct geometric and optimization-based analyses of adversarial behavior.

4.1.2. Decision boundaries and attack objective

In a multiclass classification problem with classes $C_1, C_2, \dots, C_k$, each class is represented by a discriminant function $g_i(\cdot)$. The classifier assigns a label to an input x according to:

$$\hat{y}(x) = \arg \max_i g_i(x) \quad (4)$$

To force misclassification toward the target class C_t , the perturbed input x must satisfy:

$$g_t(x) \geq g_j(x), \quad \forall j \neq t \quad (5)$$

or equivalently:

$$\max_{j \neq t} \{g_j(x)\} - g_t(x) \leq 0 \quad (6)$$

The attacker must therefore find a perturbed input x satisfying this inequality while keeping x as close as possible to the original input x_0 .

4.1.3. Optimization-based formulations

Depending on the attacker's objective and constraints, adversarial attacks can be formulated as optimization problems. Three major formulations are commonly used in the literature.

Definition 3 (Minimum-Norm Attack). This formulation seeks the smallest possible perturbation that causes misclassification:

$$\min_x \|x - x_0\| \quad \text{s.t.} \quad \max_{j \neq t} \{g_j(x)\} - g_t(x) \leq 0 \quad (7)$$

where $\|\cdot\|$ denotes a chosen norm such as ℓ_2 , ℓ_∞ , or ℓ_1 . This approach identifies the minimal change required to fool the classifier.

Definition 4 (Maximum-Allowable Attack). Here, the perturbation magnitude is bounded by a predefined threshold $\eta > 0$:

$$\min_x \max_{j \neq t} \{g_j(x)\} - g_t(x) \quad \text{s.t.} \quad \|x - x_0\| \leq \eta \quad (8)$$

This formulation ensures that the perturbation remains within an acceptable perceptual or physical range.

Table 1
Summary of deep learning approaches for EEG classification.

Author (s)	Year	Accuracy (%)	Paradigm	Dataset	Classifier type
Convolutional Neural Networks (CNNs)					
Zhao et al. [50]	2023	73.62	Motor imagery	BCI Competition IV-2a	CNN
Liu et al. [51]	2023	83.92	Motor imagery	BCI Competition IV-2a	CNN
Vega et al. [52]	2022	98.60	P300	Private P300 dataset	CNN
Han et al. [53]	2022	83.00	Motor imagery	BCI Competition IV-2b	CNN
Ak et al. [54]	2022	92.59	Motor imagery	BCI Competition IV-2a	CNN
Zhang et al. [55]	2022	62.70	Motor imagery	OpenBMI	CNN
Ma et al. [56]	2021	74.90	Motor imagery	BCI Competition IV-2a	CNN
Musallam et al. [57]	2021	83.73	Motor imagery	BCI Competition IV-2a	CNN
Vishnupriya et al. [58]	2021	84.46	Motor imagery	BCI Competition IV-2a	CNN
Shajil et al. [59]	2020	86.41	Motor imagery	Private	CNN
Miao et al. [60]	2020	90.00	Motor imagery	BCI Competition III-4a	CNN
Alazrai et al. [61]	2019	73.70	Motor imagery	Private	CNN
Korhan et al. [62]	2019	93.75	Motor imagery	BCI Competition III-3a	CNN
Transfer Learning (TL)					
Arunabala et al. [47]	2022	94.06	Motor imagery	BCI Competition IV-2a	Transfer Learning
Chen et al. [63]	2022	96.00	Motor imagery	Private	Transfer Learning
Zhang et al. [64]	2021	84.19	Motor imagery	GigaDB	Transfer Learning
Wei et al. [65]	2021	81.80	Motor imagery	BCI Competition IV-2a	Transfer Learning
Limpiti et al. [49]	2021	95.03	Motor imagery	BCI Competition IV-2a	Transfer Learning
Wei et al. [66]	2021	93.43	Motor imagery	BCI Competition II-3	Transfer Learning
Zhang et al. [67]	2021	80.00	Motor imagery	BCI Competition IV-2a	Transfer Learning
Solorzano et al. [68]	2021	74.00	Motor imagery	BCI Competition IV-2a	Transfer Learning
Deep Neural Networks (DNNs)					
Li et al. [69]	2024	80.26	Motor imagery	BCI Competition IV-2a	DNN
Wahab et al. [70]	2024	79.00	SSVEP	AVI-SSVEP dataset	VMD-DNN
Suhaimi et al. [71]	2022	49.50	Motor imagery	BCI Competition IV-2b	DNN
Cheng et al. [72]	2018	71.50	Motor imagery	Private	DNN
Other architectures					
Wei et al. [46]	2023	98.76 (98.59)	Emotion recognition	DEAP (DREAMER)	Transformer capsule network
Hwang et al. [73]	2023	97.00	Motor imagery	BCI Competition IV-2a	LSTM Autoencoder (AE)
Autthasan et al. [74]	2022	70.09	Motor imagery	BCI Competition IV-2a	LSTM Autoencoder (AE)
Saputra et al. [75]	2019	49.65	Motor imagery	BCI Competition IV-2a	LSTM Autoencoder (AE)
Han et al. [76]	2021	79.54	Motor imagery	BCI Competition IV-2a	Contrastive Learning
Ha et al. [77]	2019	77.00	Motor imagery	BCI Competition IV-2b	Capsule network
Xu et al. [78]	2020	78.50	Motor imagery	BCI Competition IV-2a	Boltzmann machine
Li et al. [79]	2016	93.57	Motor imagery	BCI Competition II-3	Deep Belief Network

Definition 5 (Regularization-Based Attack). An unconstrained optimization formulation combines the perturbation size and the attack objective into a single cost function:

$$\min_x \|x - x_0\| + \lambda \left(\max_{j \neq i} \{g_j(x)\} - g_i(x) \right) \quad (9)$$

where $\lambda > 0$ is a regularization parameter balancing the trade-off between perturbation magnitude and attack effectiveness.

The geometry of the classifier's decision boundaries plays a central role in determining its susceptibility to adversarial perturbations. In high-dimensional input spaces, even minute perturbations can move an input across a nearby decision boundary, leading to confident yet incorrect predictions. The chosen norm (e.g., ℓ_2 , ℓ_∞) defines the shape of the perturbation constraint and thus influences the type of attack generated.

Although the additive perturbation model is the most widely studied, other forms of attacks exist, such as geometric transformations, signal-space distortions, or data-poisoning attacks. Understanding these mechanisms and developing robust defense strategies against them remains an active and critical area of research in adversarial machine learning.

4.2. Adversarial attacks taxonomy

To better understand and categorize adversarial attacks, they can be classified based on the following taxonomies (see Fig. 5):

4.2.1. Specificity

- **Untargeted Attacks:** The objective is to cause the model to misclassify the input, regardless of the incorrect label it chooses. Here, the adversary aims to increase the general error rate of the model.
- **Targeted Attacks:** The goal is more specific; the adversary aims to manipulate the model into predicting a particular incorrect label. This requires a more precise perturbation that steers the model's prediction towards a chosen target class.

4.2.2. Knowledge

- **White-Box Attacks:** In this scenario, the adversary has complete access to the target model, including its architecture, parameters, and gradients. This allows for precise crafting of adversarial examples by directly optimizing the perturbations using gradient-based methods.
- **Black-Box Attacks:** Here, the adversary has limited or no knowledge of the model's internal workings. The attack must be carried out by querying the model and observing the outputs, often requiring the use of surrogate models or transferability of adversarial examples to succeed.
- **Gray-Box Attacks:** This is an intermediate scenario where the attacker has partial knowledge about the model, such as the architecture, but not the exact parameters, leading to strategies that combine elements from both white-box and black-box attacks.

Table 2 summarizes the adversarial knowledge spectrum and representative methodologies considered in security research.

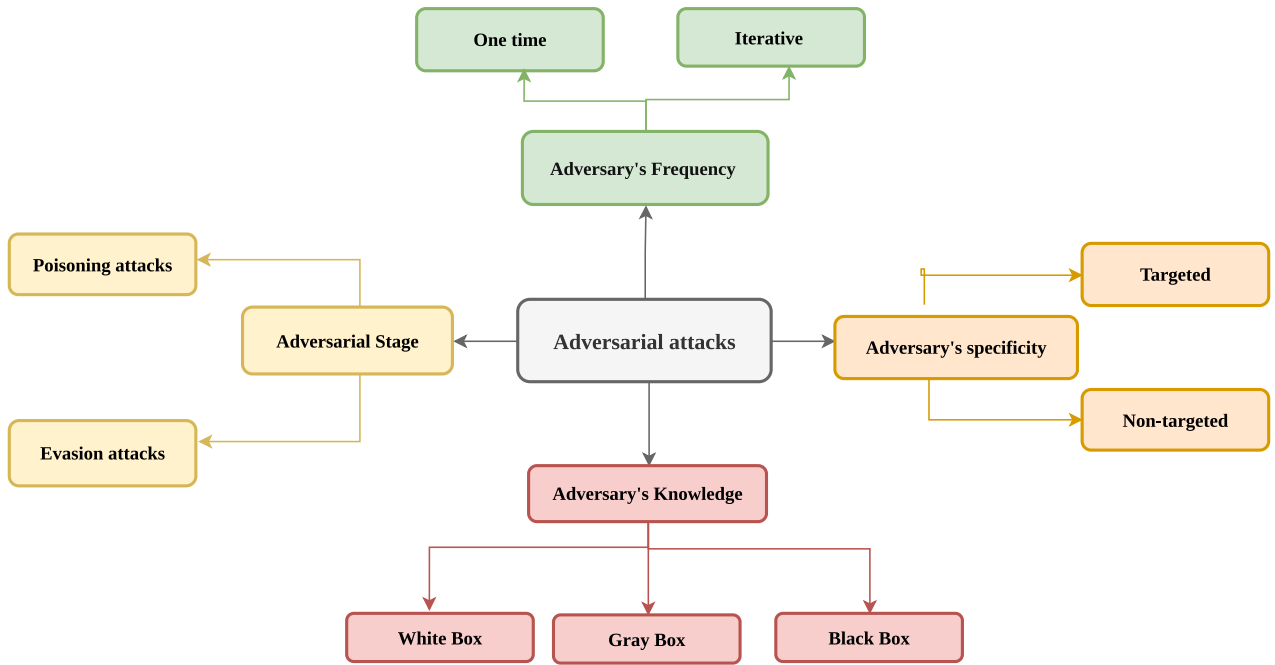


Fig. 5. Multi-dimensional taxonomy of adversarial attacks.

Table 2
Adversarial attack knowledge spectrum and methodologies.

Scenario	Adversary knowledge	Complexity/Time	Adversary attack steps
White-box attack	Full knowledge: model architecture, weights, parameters, and training data are known.	Low	1. Compute the loss gradient w.r.t. the input signal. 2. Generate the perturbation (e.g., FGSM, PGD). 3. Add the perturbation to the input to create an adversarial example.
Gray-box attack	Limited knowledge: model architecture/data partially known; some internal layer outputs may be observable.	Medium	1. Reconstruct or approximate the model using observable information. 2. Treat the reconstruction as a local white-box model. 3. Generate perturbations using the local model's gradients. 4. Transfer the perturbation to the target model.
Black-box attack	Zero knowledge: only final predictions/scores are observable (API access).	High	1. Query the target model to collect input-output pairs. 2. Train a surrogate (white-box) model from the queries. 3. Generate perturbations using the surrogate model. 4. Transfer the perturbation to the original black-box model.

4.2.3. Frequency

- **Single-Step Attacks:** The adversarial example is generated in a single iteration, such as in the Fast Gradient Sign Method (FGSM). These attacks are computationally efficient, but may be less effective against well-defended models.
- **Iterative Attacks:** These involve multiple steps to iteratively refine the adversarial perturbation, as seen in attacks like Projected Gradient Descent (PGD). Iterative methods tend to produce stronger adversarial examples at the cost of increased computational complexity.

4.2.4. Attack stages

Adversarial attacks can be categorized based on the stage at which they are executed: evasion attacks (during model testing) and poisoning attacks (during model training). The type of attack depends on how the adversary introduces adversarial perturbations:

- **Poisoning Attack:** Poisoning attacks target the training phase by injecting adversarially crafted samples into the training dataset. The goal is to subtly alter the training data, leading the resulting model to develop vulnerabilities that can be exploited later. These attacks are particularly insidious as they compromise the model from its foundation, embedding flaws that are often difficult to

detect and mitigate. They are also less prone to occur in practice. This is because they require a high degree of privilege, specifically direct access to the training pipeline, which presents a significant barrier for most potential adversaries

- **Evasion Attack:** In evasion attacks, the adversary assumes the trained model's parameters remain unchanged. The attacker generates adversarial examples that mislead the model during the testing phase. Unlike poisoning attacks, these attacks do not involve tampering with the training data or retraining the model. Instead, they manipulate the test inputs to exploit weaknesses in the model's decision making process without altering its learned parameters.

4.3. Adversarial defenses taxonomy

As adversarial attacks have gained prominence, numerous defense strategies have emerged to enhance the security of machine learning and deep learning systems. This section provides an in-depth overview of these defense mechanisms.

- (1) **Definition:** Adversarial defense encompasses a range of strategies designed to strengthen ML and DL models against adversarial attacks. The primary goal is to enhance model resilience

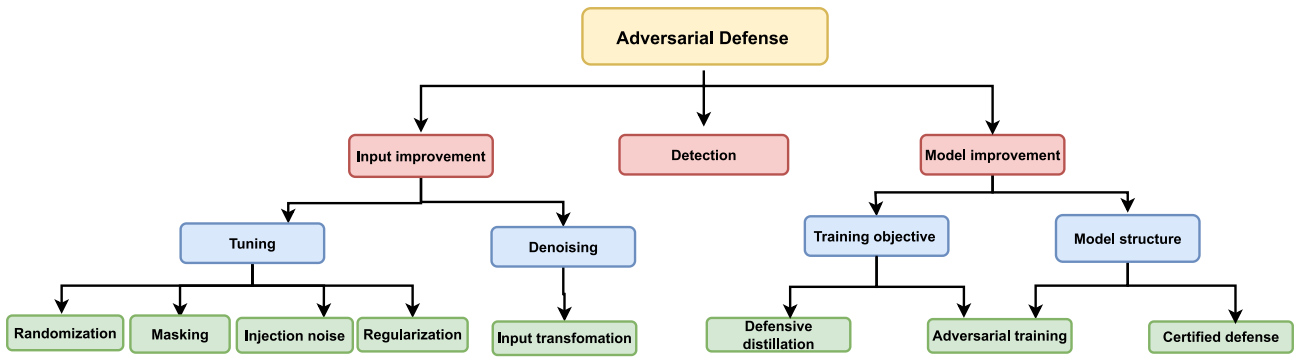


Fig. 6. Taxonomy of adversarial defense strategies for enhancing model robustness.

by preventing adversarial manipulation and ensuring consistent and accurate predictions. These defenses are critical in scenarios where models are targeted by malicious attacks. Notable defense techniques include adversarial training, preprocessing of inputs, model regularization, and defensive distillation, each addressing specific aspects of model robustness.

- (2) **Classification:** Fig. 6 outlines the main categories of adversarial defense strategies. The following subsections elaborate on these approaches in detail.

4.3.1. Model-centric adversarial defenses

Given the susceptibility of ML models to adversarial inputs, strategies aimed at improving model robustness are indispensable. Prominent approaches include adversarial training, defensive distillation, certified defense methods, and model randomization:

- **Adversarial Training:** Adversarial training represents a prominent defense strategy against adversarial attacks, involving the augmentation of the original training dataset with carefully crafted adversarial examples. By training the model to correctly classify both clean and perturbed inputs, this approach encourages smoother decision boundaries and improved resilience to small, worst-case perturbations.
 - **Single-step adversarial training:** This method typically employs gradient-based attacks, such as FGSM or single-step PGD, to generate adversarial examples during training. While computationally efficient, single-step training primarily addresses relatively simple attacks. A key limitation is the phenomenon of catastrophic overfitting, where the model appears robust to the specific adversarial examples used during training but remains vulnerable to stronger, multi-step, or adaptive attacks. Moreover, excessive focus on adversarial samples may negatively impact performance on clean inputs.
 - **Multi-step adversarial training:** Multi-step adversarial training leverages iterative attacks, such as multi-step PGD, to generate more challenging adversarial examples. This approach generally enhances robustness relative to single-step training and mitigates catastrophic overfitting. However, it incurs greater computational cost and may still exhibit robust overfitting, where training robustness does not fully generalize to unseen or adaptive attacks. While multi-step adversarial training strengthens model resilience, its effectiveness is often attack-specific, akin to a targeted immunization strategy: robustness is most pronounced against perturbations similar to those observed during training but may decline against novel attack strategies.
 - **GAN-based adversarial training:** Recent approaches utilize Generative Adversarial Networks to simulate realistic, data-driven adversarial perturbations. Unlike conventional

gradient-based methods constrained by norm bounds, GANs can generate perturbations that preserve the intrinsic characteristics of input data while inducing model misclassification. Training classifiers with GAN-generated adversarial examples enables the model to learn more invariant and discriminative feature representations.

- **Defensive Distillation:** By transferring knowledge from a complex model to a simpler one post-training, this method increases resistance to gradient-based attacks. However, it is less effective against gradient-free or approximate-gradient attacks, such as those outlined in [80], which can bypass this defense in certain black-box settings.
- **Certified Defense Methods:** These approaches introduce controlled noise into models, creating perturbations that are either below a specific threshold or linked to defined features. This strategy enhances robustness by making models less susceptible to adversarial manipulations.
- **Model Randomization:** By incorporating randomness into model operations, this technique complicates an adversary's ability to generate effective perturbations [81]. This randomness obscures gradients, reducing the reliability of gradient-based attack methods.

4.3.2. Input-centric adversarial defenses

Input-centric defense strategies aim to enhance the robustness of machine learning models by improving the quality and integrity of input data prior to or during model processing. Since model predictions are inherently dependent on input reliability, mitigating adversarial perturbations at the data level constitutes a fundamental line of defense. These approaches primarily seek to suppress, correct, or neutralize malicious modifications embedded in the input signal. Common techniques include:

- **Denoising:** Denoising-based defenses treat adversarial perturbations as a form of structured noise and attempt to suppress them through filtering or learned denoising mechanisms. The objective is to train models to become insensitive to small, adversarially induced distortions while preserving task-relevant information. Unlike adversarial detection methods, which explicitly identify and flag malicious inputs before classification, denoising techniques operate by directly attenuating perturbations within the signal itself.
- **Input Reconstruction:** Input reconstruction methods aim to restore corrupted inputs to a clean or near-clean state, typically at inference time. Rather than modifying the training process, these approaches attempt to reverse the adversarial perturbation by reconstructing the underlying signal distribution. Techniques such as defense perturbation [82] and CIIDefence [83] illustrate reconstruction-based strategies that mitigate adversarial

effects through pixel-level and feature-level restoration, thereby reducing the impact of malicious modifications on downstream classification.

- **Input Transformation:** In contrast to reconstruction, input transformation methods apply predefined or learned transformations to alter the input representation in a manner that weakens adversarial perturbations. These transformations may include signal processing operations, randomization techniques, or domain-specific preprocessing steps. Hybrid strategies, such as the approach proposed by Qin and Yue [84], combine transformation and reconstruction mechanisms to provide enhanced robustness across diverse attack scenarios.

4.3.3. Adversarial detection techniques

Detection-based defense mechanisms aim to identify adversarial inputs prior to their propagation through the machine learning (ML) model, thereby preventing compromised predictions. Rather than modifying the training process or preprocessing the input, these approaches introduce an additional verification stage that distinguishes maliciously perturbed samples from legitimate data. Broadly, adversarial detection techniques can be categorized as follows:

- **Feature-based Detection:** This class of methods examines statistical or structural patterns within the feature space to differentiate adversarial examples from clean inputs. By analyzing deviations in learned representations, activation distributions, or intermediate layer responses, these techniques exploit the fact that adversarial samples often exhibit abnormal feature characteristics compared to genuine data [85].
- **Invariant-based Detection:** Invariant-based approaches identify adversarial perturbations by leveraging features that remain stable across natural variations of the input data. These methods assess the consistency of model predictions or internal representations under controlled transformations or perturbations. Since adversarial examples frequently disrupt such invariances, deviations from expected consistency patterns can serve as reliable indicators of malicious manipulation [86].

5. How BCI can be hacked by adversarial attacks

Fig. 7 shows an EEG processing pipeline with an adversarial jamming module inserted after preprocessing and before feature extraction. The perturbed signal then propagates through the remaining stages of the pipeline without being eliminated by conventional preprocessing mechanisms. The adversarial perturbation δ can be generated using different methodological strategies. One common approach relies on gradient-based attacks, such as the Fast Gradient Sign Method (FGSM) detailed in Algorithm 1. Specifically, the attack computes the gradient of the loss function with respect to the input signal and constructs a bounded perturbation that maximizes classification error. In its single-step form (FGSM), the perturbation is obtained by adding a scaled sign of the input gradient. Iterative variants such as PGD further refine this process through repeated gradient updates combined with projection onto the ℓ_p constraint set. Although δ is constrained in magnitude to preserve physiological plausibility and remain indistinguishable from natural EEG variability, it is explicitly optimized to cross the classifier's decision boundary. Owing to the high nonlinearity of modern deep learning models, even small perturbations can induce significant shifts in the learned feature representation. Consequently, adversarially perturbed trials remain visually and spectrally similar to clean signals, yet lead to controlled misclassification. Importantly, this behavior differs fundamentally from robustness to random noise. While a BCI system may tolerate stochastic signal variations, it can still be systematically misled by structured, optimization-driven perturbations.

Despite adversarial attacks on EEG-based BCIs are well-studied, their real-world feasibility depends on system configuration and adversary access. Training-time poisoning is difficult in controlled environments, but it may occur in federated learning setups [87]. Inference-time evasion attacks require either model knowledge or proximity to acquisition hardware. These considerations contextualize assumptions about attack practicality across both phases of BCI operation.

Algorithm 1 FGSM Adversarial Attack Procedure

Input: Input sample x , target model f , true label y , perturbation budget ϵ , maximum iterations T , step size α

Output: Adversarial example x_{adv}

- (1) Initialize $x_{\text{adv}} \leftarrow x$
 - (2) **for** $t = 1$ **to** T **do**
 - (a) Compute the gradient of the loss function J with respect to x :

$$\text{gradient} \leftarrow \nabla_x J(f(x), y_{\text{true}}),$$
 where y_{true} is the true label of x .
 - (b) Compute the sign of the gradient:

$$\text{sign} \leftarrow \text{sign}(\text{gradient})$$
 - (c) Create the perturbed input by adding the scaled sign of the gradient to the original input:

$$x_{\text{perturbed}} \leftarrow x + \epsilon \cdot \text{sign}$$
 - (d) Use the perturbed input $x_{\text{perturbed}}$ to make a prediction with the target model f .
 - (3) **return** $x_{\text{perturbed}}$
-

6. Adversarial attacks in BCI

Table 3 summarizes existing studies on adversarial attacks in BCI systems, categorized by stage, adversary knowledge, specificity, frequency, and novelty. The literature indicates that EEG-based BCIs are vulnerable to both training-time and inference-time manipulations, with evasion attacks during deployment receiving the greatest attention.

6.1. Attack stage

Most studies focus on evasion attacks at the testing phase. Notable examples include CNN-based BCI vulnerabilities [88], query-efficient black-box attacks [89], imperceptible perturbations [90], and physically constrained manipulations [91]. Fewer studies investigate training-time poisoning, such as the explainability-driven approach proposed by Zhang et al. [92], which alters training data to compromise model performance.

6.2. Adversary knowledge

White-box attacks dominate the literature, enabling gradient-based methods such as PGD [93,94]. Gray-box strategies exploit partial model knowledge or transferability [95,96], whereas black-box attacks rely on query-efficient techniques [89].

6.3. Specificity and frequency

Both targeted and non-targeted strategies have been applied [88, 89,91,93]. Perturbations may be single-step or iterative, with iterative methods preferred for their higher success rates and lower visibility [97,98].

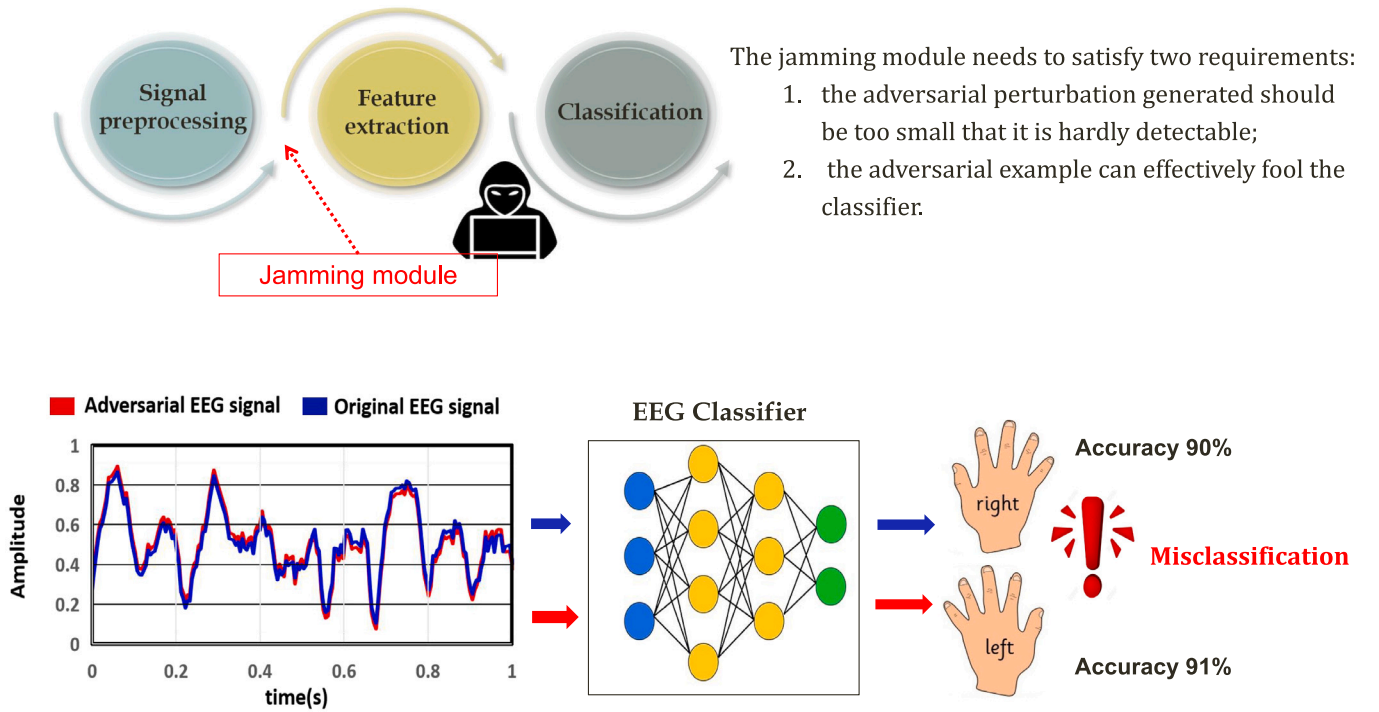


Fig. 7. Adversarial example generation pipeline and its impact on EEG classification. The original EEG signal (blue) is processed through preprocessing, feature extraction, and classification stages. The jamming module generates a constrained adversarial perturbation using a gradient-based optimization method and injects it into the signal after preprocessing. Although the perturbation produces only minor amplitude variations (red signal), it shifts the extracted feature representation across the classifier's decision boundary. As a result, the classifier outputs an incorrect label with high confidence. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3

Summary of adversarial attack works in the context of Brain–Computer Interface.

Ref.	Year	Stage		Knowledge			Specificity		Frequency		Key novelty
		Evasion	Poison	White	Gray	Black	Targeted	Non-targeted	One-time	Iterative	
[88]	2018	✓		✓	✓	✓		✓	✓	✓	First comprehensive vulnerability analysis of EEG based CNN classifiers
[89]	2019	✓				✓		✓	✓		Query-efficient black-box attack using iterative active learning
[93]	2019	✓		✓			✓			✓	White-box target attack for EEG based BCI regression problems
[90]	2021	✓		✓			✓	✓	✓		Tiny noise attacks demonstrating imperceptible perturbations
[95]	2021	✓			✓			✓		✓	Universal adversarial perturbations across subjects and models
[96]	2023	✓			✓			✓	✓	✓	Generative perturbation network for universal attacks
[94]	2023	✓		✓			✓			✓	Adversarial attacks on epilepsy diagnosis models using PGD
[22]	2024	✓		✓				✓	✓	✓	Robustness assessment of attention-based networks
[98]	2023	✓		✓				✓		✓	Frequency domain channel-wise attacks on EEG components
[99]	2024	✓		✓				✓	✓	✓	Vulnerability assessment of Capsule Network models
[91]	2022	✓		✓			✓			✓	Physically-constrained attacks with real-world signal injection
[92]	2023		✓		✓		✓			✓	Explainable data poison attacks using XAI for feature selection
[100]	2023	✓		✓				✓		✓	Gradient descent attacks on SVM classifiers for clinical EEG
[97]	2025	✓		✓				✓		✓	Attention-guided universal perturbations focusing on salient features

Table 4
Summary of adversarial defense works in the context of Brain–Computer Interface.

Reference number	Year	Detection	Input improvement		Model improvement		Key insights
			Tuning	Denoising	Training objective	Model structure	
[101]	2022	✓	N/A	N/A	✓	N/A	Evaluation of five adversarial training defense methods across three large EEG datasets, three classifiers, and multiple white-box attacks.
[21]	2023	✓	N/A	N/A	✓	N/A	Hierarchical deep learning for robust detection of adversarial attacks.
[102]	2023	N/A	✓	✓	✓	✓	Comprehensive evaluation of multiple classical and state-of-the-art adversarial defense methods for EEG based BCI.
[103]	2024	✓	✓	✓	N/A	N/A	Adversarial signals are detected and removed using ICA in the time and frequency domains, after which purified signals are reconstructed, and channel consensus voting is applied to enhance robustness.
[104]	2024	✓	N/A	N/A	N/A	N/A	Adversarial sample detection based on kernel density estimation for EEG based BCIs.
[105]	2024	✓	N/A	N/A	N/A	N/A	An automatic, frequency-based method is proposed to detect whether unseen EEG samples are normal or adversarial.
[106]	2024	✓	N/A	N/A	✓	N/A	An alignment-based adversarial training (ABAT) method is introduced, combining EEG data alignment with adversarial training to reduce domain discrepancies and strengthen classification boundaries.
[107]	2024	N/A	N/A	N/A	✓	N/A	Co-training with human EEG for limited adversarial robustness gains.
[108]	2024	N/A	N/A	✓	N/A	N/A	A denoising convolutional autoencoder (DCAE) is proposed as a preprocessing defense to remove adversarial perturbations without altering the classifier's structure.
[109]	2024	✓	N/A	N/A	N/A	N/A	The paper introduces the first adversarial detection framework using Mahalanobis and cosine distance methods to effectively identify white-box attacks.
[110]	2024	N/A	N/A	N/A	✓	N/A	Improving EEG model robustness against unseen attacks using GANs.
[111]	2024	N/A	N/A	N/A	✓	N/A	Applied adversarial training techniques to neural networks used in brain-to-brain communication systems.
[112]	2024	✓	✓	N/A	N/A	N/A	Time-domain attacks and exploration of potential defenses.
[113]	2025	N/A	✓	N/A	✓	N/A	This study shows that combining data alignment, augmentation, and robust training significantly improves both accuracy and resilience against adversarial attacks.
[114]	2024	✓	N/A	N/A	N/A	N/A	This paper proposes adversarial filtering based evasion and backdoor attacks to EEG based BCIs.
[115]	2025	N/A	✓	N/A	✓	N/A	Robustness to adversarial examples is improved and accuracy on clean images is preserved, without requiring prior knowledge of specific attacks.
[116]	2025	N/A	N/A	N/A	N/A	✓	This study explores model compression by applying structured weight pruning and quantization-aware training.
[117]	2025	N/A	N/A	N/A	✓	N/A	Adversarial training for securing SSVEP-based brain-to-brain communication.
[118]	2025	N/A	N/A	N/A	N/A	N/A	A GAN-driven defense and transfer learning strategy jointly improve adversarial detection, accuracy, and generalization across unseen attacks.

7. Adversarial defenses in BCI

Table 4 provides an overview of defense strategies proposed to counter adversarial threats in EEG based BCI systems. These approaches can be broadly categorized into **detection-based defenses, input-level improvements, and model-level improvements**. While existing works demonstrate encouraging robustness gains, the field remains in its infancy, with limited standardization and practical validation.

7.1. Detection-based defenses

Several studies have focused on detecting adversarial perturbations before they affect classification outcomes. Zhang et al. [104] proposed a dedicated adversarial sample detection framework for EEG based BCIs, while Aissa et al. [21] designed a hierarchical deep learning approach

capable of distinguishing clean from adversarial inputs. Similarly, Ding et al. [108] emphasized **multi-model evaluations across different threat models**, and Yi et al. [112] explored temporal characteristics to enhance adversarial detection. These works highlight the feasibility of filtering adversarial perturbations at the input stage, but detection-only defenses may not generalize across unseen or adaptive attacks.

7.2. Input improvement strategies

Defenses targeting the EEG signal itself have received significant attention. Yi et al. [103] applied independent component analysis (ICA)-based purification to denoise time-frequency representations of SSVEP signals. Frequency-domain defenses have also been investigated: Zhang et al. [105] and Meng et al. [102] demonstrated that perturbations exhibit distinctive spectral signatures that can be mitigated

through frequency filtering. Time-domain defenses, such as adaptive tuning approaches [109,112], also attempt to suppress adversarial noise by signal reconstruction. While input-level defenses are attractive for their model-agnostic applicability, their robustness may diminish under adaptive attacks that exploit the same preprocessing pipeline.

7.3. Model improvement strategies

Model-level defenses include modifications to training objectives and network architectures. Adversarial training has been widely explored, with Ahmadi et al. [117] applying it to secure SSVEP-based brain-to-brain communication, while Chen et al. [106] introduced alignment mechanisms to counteract gradient-based perturbations. Li et al. [101] integrated active learning to reduce query efficiency in black-box settings. Structural defenses include hierarchical or compressed network designs, such as the robust deep learning framework of Aïssa et al. [21], and the model compression strategy of Bohachenko et al. [116], which mitigates filter-level vulnerabilities by reducing redundancy. Guo et al. [107] explored co-training with human EEG to improve resilience, though with limited robustness gains. Collectively, these approaches highlight that architectural modifications and tailored training objectives can enhance robustness, but they often incur trade-offs in accuracy and computational cost.

8. Adversarial robustness evaluation

Table 5 summarizes the datasets, victim models, and evaluation metrics employed in recent studies on adversarial robustness in BCI systems. The current body of work demonstrates significant diversity in experimental design, yet several methodological gaps remain.

8.1. Datasets

A broad spectrum of EEG datasets has been utilized, spanning classical paradigms such as P300, steady-state visually evoked potentials, error-related negativity, and motor imagery, as well as affective (e.g., DEAP, AMIGOS) and clinical datasets (e.g., CHB-MIT, CAP Sleep Database). Benchmark datasets from the BCI Competitions (e.g., IV-2a, IV-2b, III-IVa) remain dominant, supporting reproducibility and comparability across studies. Nevertheless, the reliance on controlled, offline datasets limits validity. The integration of large-scale, multimodal, and naturalistic datasets—particularly those combining EEG with additional physiological modalities—remains an open challenge for evaluating adversarial robustness under realistic conditions.

8.2. Victim models

Most robustness evaluations target deep learning models specifically designed for EEG, including EEGNet, DeepConvNet, ShallowCNN, and more recently, attention-based architectures (e.g., Attention-EEGNet, TSeption, TIDNet) and capsule networks. Classical approaches such as SVM, LDA, and Ridge Regression are primarily considered in poisoning or regression tasks, while ensemble methods (e.g., Random Forest, XGBoost) have been investigated in explainability-driven adversarial settings. Despite the dominance of CNN-based frameworks, robustness assessments of emerging architectures, such as graph neural networks and transformer-based models, remain limited. Given their increasing adoption in state-of-the-art BCI pipelines, systematic adversarial evaluations of these architectures are necessary.

8.3. Evaluation metrics

Evaluation metrics in adversarial BCI research are designed to assess both model performance and the strength of adversarial perturbations. These metrics can be grouped into:

- Performance-based measures (e.g., accuracy, kappa),
- Attack effectiveness measures (e.g., success rate, fooling rate, distortion),
- Regression or signal-based metrics (e.g., mean squared error).

8.3.1. Performance-based measures

Raw Classification Accuracy (RA) quantifies the proportion of correctly classified EEG samples:

$$RA = \frac{TP + TN}{TP + TN + FP + FN}, \quad (10)$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively.

Balanced Accuracy (BA) is used to mitigate class imbalance, defined as:

$$BA = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right). \quad (11)$$

Accuracy Drop (ΔA) is employed to evaluate the degradation under attack:

$$\Delta A = A_{\text{clean}} - A_{\text{adv}}, \quad (12)$$

where A_{clean} and A_{adv} are accuracies on clean and adversarial inputs, respectively.

Cohen's Kappa (κ) measures classification consistency beyond chance:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad (13)$$

where p_o is the observed agreement and p_e is the expected agreement by random chance.

8.3.2. Attack effectiveness metrics

Attack Success Rate (ASR) quantifies the percentage of adversarial examples that successfully cause misclassification:

$$ASR = \frac{N_{\text{successful}}}{N_{\text{total}}} \times 100\%. \quad (14)$$

Fooling Rate (FR) measures the proportion of inputs for which the model's prediction changes after perturbation:

$$FR = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[f(x_i) \neq f(x'_i)], \quad (15)$$

where $\mathbb{I}[\cdot]$ is the indicator function, and x'_i is the adversarial counterpart of x_i .

Distortion (D) quantifies perturbation magnitude, often measured by the ℓ_p norm:

$$D = \|x' - x\|_p, \quad p \in \{1, 2, \infty\}. \quad (16)$$

8.3.3. Regression and signal-based metrics

For regression-oriented EEG tasks (e.g., vigilance or workload estimation), **Mean Squared Error (MSE)** and **Mean Output (MO)** are commonly employed:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \quad (17)$$

$$MO = \frac{1}{N} \sum_{i=1}^N \hat{y}_i, \quad (18)$$

where y_i and $\hat{y}_i$ are the ground-truth and predicted outputs.

8.3.4. Precision-based metrics

In poisoning and explainability-based attacks, performance is assessed through precision, recall, and F1-score:

$$\text{Precision} = \frac{TP}{TP + FP} \quad \text{Recall} = \frac{TP}{TP + FN} \quad (19)$$

$$F1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (20)$$

Table 5
Experimental setup and evaluation metrics used in BCI adversarial security research.

Reference	Dataset	Victim model	Metrics
[93]	1. Driving dataset 2. Psychomotor Vigilance Task dataset	1. Ridge Regression (RR), 2. MLP	1. Mean Squared Error (MSE), 2. Mean Output (MO), 3. Attack Success Rate, 4. Distortion.
[89]	1. P300 evoked potentials, 2. BCI Competition IV-2a, 3. ERN dataset	1. EEGNet, 2. DeepConvNet, 3. ShallowCNN	1. Raw classification accuracy, 2. Balanced accuracy.
[90]	1. P300 dataset [119], 2. SSVEP dataset [120]	Riemannian geometry model	1. Accuracy, 2. Fooling rate.
[95]	1. P300 evoked potentials, 2. BCI Competition IV-2a, 3. ERN dataset	1. EEGNet, 2. DeepConvNet, 3. ShallowCNN	1. Raw classification accuracy, 2. Balanced accuracy.
[76]	BCI Competition IV-2a	Semi-supervised CNN	1. Robust accuracy, 2. Fooling rate.
[98]	1. BCI Competition IV-2a, 2. BCI Competition IV-2b, 3. BCI Competition III-IVa, 4. Motor Imagery 2019	1. EEGNet, 2. DeepConvNet, 3. ShallowCNN, 4. LDA, 5. SVM, 6. LSTM, 7. GRU	1. Accuracy, 2. Computational time.
[96]	1. AMIGOS [121], 2. DEAP [122], 3. NER2015 [123], 4. Physionet [124]	1. EEGNet, 2. DeepConvNet, 3. ShallowCNN, 4. TIDNet, 5. ResNet, 6. VGG	1. Average accuracy, 2. Fooling rate.
[94]	CHB-MIT Scalp [125]	1. Maxpool temporal, 2. Convolution LSTM, 3. Mixed LSTM	1. Success rate, 2. Distortion level, 3. Accuracy.
[92]	Private dataset	1. Random Forest, 2. AdaBoost, 3. Extra Trees, 4. XGBoost, 5. MLP, 6. KNN	1. Poison rate, 2. Accuracy, 3. Recall, 4. Precision, 5. F1-score.
[22]	BCI Competition IV-2a	Attention-EEGNet	1. Accuracy drop, 2. Kappa, 3. Confusion matrix.
[99]	BCI Competition IV-2b	1. CapsNet-EEG, 2. EEGNet.	1. Accuracy, 2. Kappa.
[97]	1. P300 evoked potentials, 2. BCI Competition IV-2a, 3. ERN dataset	1. EEGNet, 2. DeepConvNet, 3. ShallowCNN, 4. TIDNet, 5. TSception	1. Accuracy success rate, 2. Raw accuracy, 3. Balanced accuracy.

8.4. Impact of adversarial robustness in EEG-based BCIs

Beyond quantitative evaluation metrics, understanding the practical implications of adversarial robustness is essential for real-world EEG-based BCI deployment. The degradation observed through metrics such as Accuracy Drop, ASR, and Distortion directly translates into tangible risks in operational environments.

(1) Assistive Communication Systems: In P300- or SSVEP-based speller systems used by patients with severe motor impairments (e.g., amyotrophic lateral sclerosis), adversarial perturbations that increase ASR may lead to incorrect character selection. Even small perturbations (low ℓ_p distortion) can cause repeated misclassification, resulting in communication delays, user frustration, and reduced trust in the system. Robustness evaluation therefore reflects the reliability of assistive communication in daily use.

(2) Motor Imagery-Based Neuroprosthetics: In motor imagery BCIs controlling robotic arms or wheelchairs, a successful adversarial attack may trigger unintended movements. For instance, a misclassification between left- and right-hand imagery could cause unsafe directional commands. Accuracy Drop (ΔA) under attack directly

corresponds to the probability of hazardous control errors, making robustness a safety-critical requirement.

(3) Clinical Monitoring and Epileptic Seizure Detection: In clinical EEG applications such as seizure detection (e.g., CHB-MIT dataset), adversarial perturbations that increase false negatives may prevent timely intervention, while false positives could lead to unnecessary alarms. Metrics such as precision, recall, and MSE in regression settings quantify these clinically relevant risks.

(4) Affective and Cognitive State Monitoring: In workload or vigilance estimation systems used in driving or aviation scenarios, adversarial manipulation may mask fatigue states or induce false alerts. Changes in regression-based metrics (e.g., MSE) reflect the reliability of cognitive-state assessment under adversarial conditions.

(5) Security and Privacy-Sensitive Environments: In authentication or intent-recognition BCIs, high FR or ASR indicates vulnerability to spoofing or impersonation attacks. Robustness evaluation therefore informs the design of secure deployment frameworks incorporating anomaly detection and adaptive defenses.

9. Research challenges and open issues

While significant progress has been made in understanding adversarial vulnerabilities in EEG based BCIs, several challenges remain unresolved. These limitations hinder both the reliability of robustness evaluations and the practical deployment of defenses. This section outlines key research challenges and open issues.

9.1. Dataset limitations and ecological validity

Most adversarial attack and defense studies rely on a small set of benchmark datasets (e.g., BCI Competition IV-2a, DEAP, CHB-MIT), which are collected under controlled laboratory conditions. Such datasets often lack ecological diversity, multi-session variability, and cross-device recordings. Consequently, defenses validated on these datasets may not generalize to real-world scenarios. A major challenge lies in developing large-scale, multimodal, and cross-subject datasets that reflect realistic usage conditions.

9.2. Standardized evaluation protocols

Current studies adopt heterogeneous evaluation criteria, including accuracy drop, fooling rate, success rate, distortion measures, and kappa scores. The absence of a unified robustness benchmark complicates fair comparison across works. Establishing standardized evaluation protocols, similar to those in computer vision, is an open challenge necessary to enable systematic progress in adversarial robustness for BCIs.

9.3. Adaptive and real time adversaries

Most research has focused on static offline adversarial examples, whereas practical BCIs operate in real time, closed-loop environments. Adaptive adversaries, capable of adjusting attacks dynamically during interaction, remain largely unexplored. Designing attacks and defenses under streaming and online conditions is a critical open issue for ensuring real-world robustness.

9.4. Defense generalization and transferability

Many existing defenses (e.g., adversarial training, signal purification, detection mechanisms) are effective only against specific attack families. Their generalization to unseen or adaptive attacks remains weak. Furthermore, the transferability of defenses across datasets, subjects, and recording devices has not been systematically evaluated. Future research must prioritize the development of generalized, adaptive, and lightweight defenses.

9.5. Trade-offs between robustness, accuracy, and efficiency

Adversarial training and model modifications often improve robustness at the expense of accuracy and computational efficiency. In resource-constrained BCI applications (e.g., wearable or implantable systems), such trade-offs are particularly problematic. A key research question is how to design defense mechanisms that balance robustness with efficiency, usability, and user safety.

9.6. Explainability and trustworthiness

Beyond performance considerations, adversarial vulnerabilities may also undermine trust in BCI technologies. If users or clinicians perceive that neural decoding systems can be easily manipulated, confidence in computer-assisted neurotechnology could be significantly compromised. Therefore, it is essential to develop robust, reliable, and interpretable BCI systems that maintain stable performance under both benign and adversarial conditions. Few works have investigated how

adversarial perturbations interact with neural representations learned from EEG. The lack of interpretability impedes trust in both attack and defense evaluations. Incorporating explainable AI (XAI) into adversarial robustness research may provide deeper insights into model vulnerabilities and improve defense design.

9.7. Security in emerging BCI paradigms

Most adversarial research has focused on classification-oriented EEG tasks. However, BCIs are rapidly expanding into affective computing, brain-to-brain communication [111], and neuroadaptive interfaces. Each of these paradigms introduces unique security challenges, such as privacy leakage and manipulation of cognitive states. Extending adversarial robustness research to these emerging domains is an open issue with significant practical implications.

9.8. Regulatory and ethical considerations

Adversarial manipulation of neural signals raises critical ethical and regulatory questions, including data privacy, informed consent, and accountability in clinical or assistive applications. For instance, unauthorized access to neural data could expose sensitive cognitive states, demanding encryption and consent protocols. In human-in-the-loop contexts, where users actively interact with systems, safety risks from signal misinterpretation – such as erroneous prosthetic movements – require fail-safes like emergency overrides and continuous human oversight to preserve autonomy and prevent harm. Regulatory frameworks classify high-risk BCIs as medical devices under FDA Class III or EU MDR rules, enforcing rigorous premarket testing, post-market surveillance, and neurosecurity standards against hijacking or breaches, with potential for international harmonization. Addressing these concerns requires interdisciplinary collaboration beyond machine learning, involving neuroscience, law, and ethics (see Table 6).

10. Conclusion

BCIs have demonstrated great potential in healthcare, assistive technologies, and human machine interaction. However, as this survey highlights, the increasing reliance on machine learning models exposes BCI systems to adversarial vulnerabilities that threaten their reliability, security, and trustworthiness. We provided a comprehensive review of adversarial attacks targeting EEG based BCIs, covering both training-time and testing-time threats, and analyzed their taxonomy in terms of adversarial knowledge, attack specificity, and perturbation strategies. We also reviewed existing defenses, including detection frameworks, input purification techniques, and model-level improvements such as adversarial training and architectural modifications.

Despite recent progress, adversarial robustness in BCI remains in early stage. Current defenses are often limited in scope, evaluated on small-scale datasets, and lack generalization to unseen or adaptive attacks. Furthermore, most studies rely on offline evaluation settings, which fall short of addressing the challenges of real time, closed-loop BCI applications. Our discussion of research challenges highlighted critical open issues related to dataset limitations, standardized evaluation protocols, defense generalization, robustness efficiency trade-offs, explainability, and ethical considerations. Looking ahead, future research should prioritize the development of large-scale, and multimodal datasets, alongside the design of standardized robustness benchmarks for fair comparison. Lightweight and adaptive defenses that balance security, accuracy, and efficiency will be crucial for real-world deployment, particularly in wearable and clinical BCIs. Moreover, explainability and ethical frameworks must be integrated into robustness research to ensure user trust and societal acceptance. By addressing these challenges, the BCI community can move toward secure, resilient, and trustworthy next-generation systems capable of supporting both scientific exploration and practical applications.

Table 6

Research challenges and open issues in adversarial robustness for BCIs and proposed solutions.

Challenge/Open issue	Description	Proposed solutions/Future directions
Dataset limitations	Reliance on small, controlled datasets (e.g., BCI Competitions, DEAP, CHB-MIT) lacking ecological validity, cross-session variability, and multimodality.	Develop large-scale, multimodal, cross-subject datasets; incorporate real-world and wearable BCI recordings; encourage open data-sharing initiatives.
Lack of standardized evaluation protocols	Metrics vary widely (accuracy drop, fooling rate, kappa, distortion), making cross-study comparisons difficult.	Establish unified benchmarks and standardized evaluation protocols for BCI robustness, similar to those in computer vision.
Adaptive and real time attacks	Most works focus on offline adversarial examples; dynamic, real time adaptive attacks are rarely studied.	Develop streaming adversarial models and evaluate defenses in closed-loop, real time BCI systems.
Defense generalization	Defenses often tailored to specific attack families, failing to generalize to unseen or adaptive adversaries.	Explore ensemble defenses combining detection, adversarial training, and input purification; design adaptive defenses resilient to diverse threat models.
Robustness–Efficiency trade-offs	Adversarial training and robust architectures improve security but reduce accuracy and increase computation.	Investigate lightweight models, knowledge distillation, pruning, and hardware-efficient defense mechanisms for wearable BCIs.
Explainability and Trustworthiness	Limited understanding of how adversarial perturbations affect learned EEG representations, reducing interpretability.	Integrate explainable AI (XAI) into robustness analysis; design interpretable defenses that provide transparency in model decisions.
Security in emerging BCI paradigms	Current focus is on classification tasks; new domains (affective BCIs, brain-to-brain communication, neuroadaptive systems) remain unexplored.	Extend adversarial robustness research to non-traditional BCI paradigms; assess risks of privacy leakage and cognitive manipulation.
Regulatory and ethical issues	Adversarial manipulation of neural signals raises privacy, consent, and accountability concerns in clinical and assistive settings.	Develop ethical guidelines and regulatory frameworks; promote interdisciplinary collaboration between AI, neuroscience, law, and ethics.

CRedit authorship contribution statement

Nour El Houda Sayah Ben Aissa: Writing – original draft, Resources, Investigation, Formal analysis, Data curation. **Chaker Abdelaziz Kerrache:** Writing – original draft, Supervision, Methodology, Investigation. **Carlos T. Calafate:** Writing – review & editing, Supervision, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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